Tribhuvan University
Institute of Science and Technology
2080

Bachelor Level / First Year / First Semester — Full Marks: 60
Computer Science and Information Technology (CSC114) — Pass Marks: 24
(Introduction to Information Technology) Model Question — Time: 3 hours

Candidates are required to give answers in their own words and as far as practicable.

Section A

Attempt any two questions. (2 × 10 = 20)

1. What is operating system? Discuss different functions of operating system in detail. (2 + 8)
2. Why do we need computer network? Discuss different types of network topologies along with their merits and demerits. (3 + 7)
3. What are the benefits of storing data using databases? Discuss three levels of database system architecture in detail. (3 + 7)

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Discuss characteristics of third generation of computers. Compare it with fourth generation. (3 + 2)
5. Discuss the components of CPU in brief. (5)
6. What is primary memory? Discuss different types of primary memory. (1 + 4)
7. Define hard-copy and soft-copy output. Differentiate between impact and non-impact printers with example. (2 + 3)
8. Convert $(AF7)_{16}$ to binary. Subtract $(1001101)_2$ from $(1100011)_2$. (2 + 3)
9. Define IP address with example. What are the benefits of using domain name? (2 + 3)
10. What are the characteristics of multimedia? Discuss. (5)
11. Define cryptography. Discuss public key cryptography in detail. (1 + 4)
12. Write short notes on: (2 × 2.5 = 5)
    a. BigData
    b. E-governance

Tribhuvan University
Institute of Science and Technology
2080

Bachelor Level / First Year / First Semester — Full Marks: 60
Computer Science and Information Technology (CSC115) — Pass Marks: 24
(C Programming) Model Question — Time: 3 hours

Candidates are required to give answers in their own words and as far as practicable.

Section A

Attempt any two questions. (2 × 10 = 20)

1. Define function and list its advantages. Describe the difference between passing arguments by value and passing arguments by address with suitable program. [4+6]
2. Explain how structure is different from union? Make a program using structure of booklist having data member's title, author, and cost. Enter four data and calculate total cost. [3+4+3]
3. Explain various modes in which file can be opened? Write a program to CREATE and WRITE N numbers in a file "NUMBER.TXT". Open this file then read its content and put all even numbers in one file "EVEN.TXT" and odd numbers in another file "ODD.TXT". [2+4+4]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. What do you mean by a problem analysis? What are the properties of a good algorithm? Explain the Compilation and Execution of any C program? [1+1+3]

5. Define nested if else statement with suitable flowchart. Write a C code to check if user given input is exactly divisible by 5 or 11 using nested if else statement? [2+3]
6. List various binary and unary operators used in C? Write a program that uses a "while" loop to compute and prints the sum of a given numbers of squares. For example, if 4 is input, then the program will print 30, which is equal to $1^2+2^2+3^2+4^2$. [1+4]
7. "Size of character array is always declared one more than the input size." Justify the statement. Write a program to read a character array input as "TRIBHUVAN UNIVERSITY" from the user and find out how many times a character 'I' occurs in that array? [1+4]

8. Write syntax to declare and initialize 2-dimensional array? With suitable program logic explain how would you find transpose of a 3*3 matrix? [1+4]

9. Explain the concept of recursive function using the example program to find the factorial of given positive integer. [5]
10. Describe the fundamental concept of pointer and its arithmetic with suitable examples. [5]
11. Explain the use of graphical functions. Write a program to draw a triangle using line() graphics function. [1+4]
12. Write short notes on: [2+3]
    i) Dynamic Memory Allocation
    ii) break and continue

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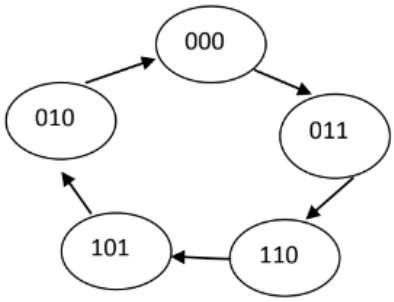
Bachelor Level / First Year / First Semester — Full Marks: 60
Computer Science and Information Technology (CSC116) — Pass Marks: 24
(Digital Logic) Model Question — Time: 3 hours

Candidates are required to give answers in their own words and as far as practicable.

Section A

Attempt any two questions. (2 × 10 = 20)

1. Implement the following function F=Σ(0,1,3,4,7) using
   a) Decoder
   b) Multiplexer
   c) PLA
2. Differentiate between synchronous and asynchronous sequential circuit. Design a counter as shown in the state diagram below



3. Explain different types of shift registers with necessary diagrams.

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Convert (2AC5)16 to decimal, octal and binary.
5. What do you mean by encoder? Design 3 to 8 line encoder.
6. Design a combinational circuit that multiplies 2-bit numbers, a1a0 and b1b0 to produce a 4-bit product, c3c2c1c0. Use AND gates and half-adders.
7. Design a circuit which produces 2's complement of the given four bit binary digit.
8. Implement full adder using decoder with truth table and logic diagram.
9. Design a circuit that produces the square of three bit number using ROM?

10. Use K-map to simplify the given function in POS. Implement the simplified function using 2-input NOR-NOR gate only.

    $F = \prod M(0,1,2,9,10,11,14)$ And with don't care conditions

    $D = \prod M(7,8,12)$

11. Discuss race condition in J-K Flip flop and methods to overcome it.
12. Write Short notes on (A**ny two**)
    a) Coding system in logic design
    b) Error-detection code
    c) Universal Gates

Tribhuvan University
Institute of Science and Technology
2080

Bachelor Level / Second Year / Fourth Semester
Computer Science and Information Technology (MTH117)
(Mathematics I) Model Question

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to give answers in their own words and as far as practicable.

Section A

Attempt any two questions. (2 × 10 = 20)

1. (a) Sketch the graph of the function $y^2 = x$ and find the domain and range of the function. [5]

   (b) (a) Dry air is moving upward. If the ground temperature is 20° and the temperature at a height of $1km$ is 10^oC, express the temperature T in ^oC as a function of the height h (in kilometers), assuming that a linear model is appropriate. (b) Draw the graph of the function in part (a). What does the slope represent? (c) What is the temperature at a height of $2km$? [5]

2. (a) A famer has $2000ft$ of fencing and wants to fence off a rectangular field that borders a straighrt river. He needs no fence along the river. What are the dimensions of the field that has the largest area? [5]

   (b) Show that the $\int_1^\infty \frac{1}{x^2}$ converges and $\int_1^\infty \frac{1}{x}$ diverges. [2]

   (c) If $f(x, y) = xy/(x^2 + y^2)$, does f(x, y) exist, as $(x, y) \to (0, 0)$? [3]

3. (a) Evaluate [5]

$$\int_{-3}^{2}\int_{0}^{\pi/2}(y + y^2 \cos x)dxdy$$

   (b) Find the Maclaurin's series for $\cos x$ and prove that it represents $\cos x$ for all x. [5]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Define continuity of a function at a point $x = a$. Show that the function $f(x) = \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.
5. State Rolle's theorem and verify the Rolle's theorem for $f(x) = x^3 - x^2 - 6x + 2$ in $[0, 3]$.
6. Find the third approximation x_3 to the root of the equation $f(x) = x^3 - 2x - 7$, setting $x_1 = 2$.
7. Find the derivative of $\mathbf{r}(t) = (1 + t^2)\mathbf{i} - te^{-t}\mathbf{j} + \sin 2t\mathbf{k}$ and find the unit tangent vector at $t = 0$.
8. Find the volume of the solid obtained by rotating about the y-axis the region between $y = x$ and $y = x^2$.
9. Solve: $y'' + y' = 0, y(0) = 5, y(\pi/4) = 3$
10. Find a vector perpendicular to the plane that passes through the points: $P(1, 4, 6), Q(-2, 5, -1)$ and $R(1, -1, 1)$
11. Find the partial derivative of $f(x, y) = x^3 + 2x^2y^3 - 3y^2 + x + y$, at $(2, 1)$.
12. Find the local maximum and minimum values, saddle points of $f(x, y) = x^4 + y^4 - 4xy + 1$.

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2080

Bachelor Level / First Year / First Semester — Full Marks: 60
Computer Science and Information Technology (PHY118) — Pass Marks: 24
(Physics) Model Question — Time: 3 hours

Candidates are required to give answers in their own words and as far as practicable.

Section A

Attempt any two questions. (2 × 10 = 20)

1. What do you mean by the equilibrium current across the *pn* junction? Use Fermi-Dirac statistics and Maxwell-Boltzmann distribution to show the flow of electrons from *n* to *p* is equal to the flow from *p* to *n*. How electron current from *p* to *n* (that is, associated with minority carriers) is not affected by the height of the potential energy barrier? Explain. [10]

2. Set up differential equation for an oscillation of a spring using Hooke's and Newton's second law. Find the general solution of this equation and hence the expressions for period, velocity and acceleration of oscillation. [10]

3. Set up Schrodinger equation for Hydrogen atom using spherical polar coordinates and separate radial and angular part of this equation. Without solving radial and angular equations, discuss the quantum numbers associated with these. [10]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Explain Hall effect and discuss the importance of Hall voltage while manufacturing electronic devices. [5]

5. What do you mean by Bloch theorem? Discuss its use in Kronig-Penny model and hence in band theory. [5]

6. Describe the following process of IC production: (a) Oxidation, (b) Pattern definition, and (c) Doping. [5]

7. A given spring stretches 0.1 m when a force of 20 N pulls on it. A 2-kg block attached to it on a frictionless surface is pulled to the right 0.2 m and released. (a) What is the frequency of oscillation of the block? (b) What is its velocity at the midpoint? (c) What is its acceleration at either end? (d) What are the velocity and acceleration when x = 0.12 m, on the block's first passing this point? [5]

8. Two large parallel plates are separated by a distance of 5 cm. The plates have equal but opposite charges that create an electric field in the region between the plates. An a particle ($q = 3.2 \times 10^{-19}$ C, $m = 6.68 \times 10^{-27}$ kg) is released from the positively charged plate, and it strikes the negatively charged plate 2×10^{-6} sec later. Assuming that the electric field between the plates is uniform and perpendicular to the plates, what is the strength of the electric field? [5]

9. What are (a) the energy, (b) the momentum, and (c) the wavelength of the photon that is emitted when a hydrogen atom undergoes a transition from the state $n = 3$ to $n = 1$? [5]

10. For a free quantum particle show that the wavefunction, $\psi(x,t) = Acoskx\, e^{-i\omega t}$ satisfies the time dependent Schrodinger equation. [5]

11. Copper has a face-centered cubic structure with a one-atom basis. The density of copper is 8.96 g/cm^3 and its atomic weight is 63.5 g/mole. What is the length of the unit cube of the structure? [5]

12. The output of a digital circuit (y) is given by this expression:

$$y = (\bar{A}B + \bar{C}A)(\overline{B + C})$$

Where A, B and C represent inputs. Draw a circuit of above equation using OR, AND and NOT gate and hence find its truth table. [5]

Tribhuvan University
Institute of Science and Technology
2080

Bachelor Level / First Year / Second Semester — Full Marks: 60
Computer Science and Information Technology (CSC165) — Pass Marks: 24
(Discrete Structure) Model Question — Time: 3 hours

Candidates are required to give answers in their own words and as far as practicable.

Section A

Attempt any two questions. (2 × 10 = 20)

1. Why breaking down of large integer into set of small integers is preferred while performing integer arithmetic? Find sum of numbers 123,684 and 413,456 by representing the numbers as 4-tuple by using reminders modulo of pair-wise relatively prime numbers less than 100. {2+8}
2. Define linear homogeneous recurrence relation. Solve the recurrence relation $a_n=a_{n/2}+n+1$, with $a_1=1$.Also discuss about probabilistic primility testing with example. {2+4+4}
3. How Zero-one matrix and diagraphs can be used to represent a relation? Explain the process of identifying whether the graph is reflexive, symmetric, or anti-symmetric by using matrix or diagraph with suitable example. {4+6}

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Prove that $\overline{A \cap B} = A \cup B$ by using set builder notation. How sets are represented by using bit string? Why it is preferred over unordered representation of sets? {3+2}
5. How can you relate domain and co-domain of functions with functions in programming language? Discuss composite and inverse of function with suitable examples. {2+3}
6. State Euclidean and extended Euclidean theorem. Write down extended Euclidean algorithm and illustrate it with example. {1+4}
7. State and prove generalized pigeonhole principle? How many cards should be selected from a deck of 52 cards to guarantee at least three cards of same suit? {2.5+ 2.5}
8. Represent the argument "If it does not rain or if is not foggy then the sailing race will be held and the lifesaving demonstration will go on. If sailing race is held then trophy will be awarded. The trophy was not awarded. Therefore it not rained" in propositional logic and prove the conclusion by using rules of inferences. {2+3}
9. Discuss common mistakes in proof briefly. Show that n is even if n^3+5 is odd by using

indirect proof. {2+3}

10. How mathematical induction differs from strong induction? Prove that $1^2 + 2^2 + 3^2 + \cdots \ldots \ldots . n^2 = n(n+1)(2n+1)/6$ by using strong induction. {1+4}
11. Write down recursive algorithm for computing a^n. Argue that your algorithm is correct by using induction. {2.5+2.5}
12. What is meant by chromatic number? How can you use graph coloring to schedule exams? Justify by using 10 subjects assuming that the pairs {(1,2), (1,5), (1,8), (2,4), (2,9), (2,7), (3,6), (3,7), (3,10), (4,8), (4,3), (4,10), (5,6), (5,7)} of subjects have common students. {1+4}

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Bachelor Level / First Year / Second Semester — Full Marks: 60
Computer Science and Information Technology (CSC166) — Pass Marks: 24
(Object Oriented Programming) Model Question — Time: 3 hours

Candidates are required to give answers in their own words and as far as practicable.

Section A

Attempt any two questions. (2 × 10 = 20)

1. What is object oriented programming? Explain objects, class, encapsulation, data hiding, inheritance, and polymorphism. [2+8]
2. Explain operator overloading. Write a program that overloads insertion and extraction operators. [2+8]
3. What is inheritance? Explain the ambiguities associated with multiple inheritance with suitable example programs. [2+8]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Explain the purpose of a namespace with suitable example.
5. What is the principle reason for passing arguments by reference? Explain with suitable code.
6. Why constructor is needed? Explain different types of constructors with example.
7. Write a program that illustrates the conversions between objects of different classes having conversion function in source object.
8. Explain the difference between private and public inheritance with suitable diagram.
9. How late binding is different from early binding. Write a program that explains late binding using virtual function.
10. Why do we need exceptions? Explain “exceptions with arguments” with suitable program.
11. What are the advantages of using the stream classes for I/O? Write a program that writes object to a file.
12. Write a program to demonstrate concept of this pointer.

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Institute of Science and Technology
2080

Bachelor Level / First Year / Second Semester — Full Marks: 60
Computer Science and Information Technology (CSC167) — Pass Marks: 24
(Microprocessor) Model Question — Time: 3 hours

Candidates are required to give answers in their own words and as far as practicable.

Section A

Attempt any two questions. (2 × 10 = 20)

1. Draw logical block diagram of 8086 microprocessor and explain its segmented memory structure.
2. What is machine cycle and instruction cycle? Draw a timing diagram for STA 2000h memory instruction. (Choose any memory locations for loading STA 2000h instruction)
3. Write an assembly language program to sort an array in ascending order using 8 bit microprocessor. (Assume appropriate array data and address where minimum array size of 10 should be considered)

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Draw pin diagram of 8085 microprocessor with appropriate labelling.
5. Specify the output in PORT1 after the execution of the following program. Write comments for each instruction.

```
MVI A, AAH
MOV B, A
RRC
XRA B
OUT PORT1
HLT
```

6. What is DMA? Explain the sequence of events that occurs during DMA operation?
7. What is addressing mode? Explain different addressing mode in 8085 microprocessor.
8. Write a program to reverse a given a string using 16 bit microprocessor.
9. Explain memory interfacing in 8085 microprocessor along with appropriate diagram.
10. What are different operating modes in 80286 microprocessor? Explain in brief about each mode.

11. “Interrupt based I/O is efficient compared to polled I/O”. Justify this statement with general working mechanism in both methods.
12. Write Short Notes (Any Two):
    a) Macro Assembler
    b) BSR Mode
    c) System Bus

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Bachelor Level / First Year / Second Semester — Full Marks: 60
Computer Science and Information Technology (MTH168) — Pass Marks: 24
(Mathematics II) Model Question — Time: 3 hours

Candidates are required to give answers in their own words and as far as practicable.

Section A

Attempt any two questions. (2 × 10 = 20)

1. When a system of equations is consistent and inconsistent? Expressing the system in the augmented matrix form determine if the system [2+8]

$$\begin{aligned} x_2 + 4x_3 &= 5, \\ 2x_1 - 3x_2 + 2x_3 &= 1, \\ 5x_1 - 8x_2 + 7x_3 &= 1 \end{aligned}$$

is consistent, by reduced echolen form.

2. Let $\mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}, \mathrm{v} = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}.\mathrm{w} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$. If the linear combination $x_1\mathbf{u} + x_2\mathbf{v} = \mathbf{w}$ exists, determine x_1 amd x_2.

3. Find the equation $y = a_0 + a_1 x$ of the least squares line that best fits the data points $(0, 1), (1, 1), (2, 2), (3, 2)$.

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Demographic studies show that each year about 3% of a city's population moves to the suburbs, while 5% of the subruban population moves to the city. Find the migration matrix. If the population of the city in 2022 was 500000 and 400000 in the subrubs, compute the population of the region in 2023.

5. When a linear system of equation is consistent? Determine whether the system: $2x_1 - x_2 = 5; 6x_1 - 3x_2 = 8$ is consistent? [1+4]

6. When two column vectors in $\mathbb{R}^2$ are equal? Give an example. Compute $4\mathrm{a} + 3\mathrm{b}, 4\mathrm{a} - 3\mathrm{b}$, where [1 + 4]

$$\mathrm{a} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}.\mathrm{b} = \begin{bmatrix} 0 \\ -1 \\ -3 \end{bmatrix}.$$

7. When two matrices A and B are commutative? Check whether the matrices

$$A = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix},$$

are commutative or not.

8. Find a matrix A such that $\mathbf{V} = Col A$, where

$$V = \left\{ \begin{bmatrix} 5\mathbf{a} - \mathbf{b} \\ \mathbf{a} - \mathbf{b} \\ -3\mathbf{a} \end{bmatrix} : \mathbf{a}, \mathbf{b} \in \mathbb{R} \right\}$$

9. Find the eigenvalues of $\begin{bmatrix} 2 & 1 \\ -3 & 6 \end{bmatrix}$. [5]

10. Find the solution of the equation [5]

$$y_{k+3} - 2y_{k+2} - 5y_{k+1} + 6y_k = 0 \quad \text{for all } k.$$

11. Define group. Show that the set of integers is not a group with respect to subtraction operation. [2+3]

12. Define ring. Show that set of real numbers with respect to addition and multiplication operation is a ring. [2+3]

Tribhuvan University
Institute of Science and Technology
2080

Bachelor Level / First Year / Second Semester — Full Marks: 60
Computer Science and Information Technology (STA169) — Pass Marks: 24
(Statistics I) Model Question — Time: 3 hours

Candidates are required to give answers in their own words and as far as practicable.

Section A

Attempt any two questions. (2 × 10 = 20)

1. A new computer program consists of two modules. The first module contains an error with probability 0.2. The second module is more complex; it has a probability of 0.4 to contain an error, independently of the first module. An error in the first module alone causes the program to crash with probability 0.5. For the second module, this probability is 0.8. If there are errors in both modules, the program crashes with probability 0.9. Suppose the program crashed. What is the probability of errors in both modules?

2. Explain how box-plot is helpful to know the shape of the data distribution. The following data set represents the number of new computer accounts registered during ten consecutive days.

43	**37**	**50**	**51**	**58**	**105**	**52**	**45**	**45**	**10**

a) Compute the mean, median, quartiles, and sample standard deviation.
b) Check whether there are outliers or not.
c) If outliers are present, then delete the detected outliers and compute the mean, median, quartiles, and sample standard deviation again.
d) Make your conclusion about the effect of outliers on descriptive statistical analysis.

3. A computer manager interested to know how efficiency of his/her new computer program which depends on the size of incoming data. Efficiency will be measured by the number of processed requests per hour. In general, larger data sets require more computer time, and therefore, fewer requests are processed within 1 hour. Applying the program to data sets of different sizes, the following data were gathered.

Data size(gigabytes)	**6**	**7**	**7**	**8**	**10**	**10**	**15**
Processed requests	40	55	50	41	17	26	16

a) Identify which one response variable, and fit a simple regression line, assuming that the relationship between them is linear.
b) Interpret the regression coefficient with reference to your problem.
c) Obtain coefficient of determination, and interpret this.
d) Based on the fitted model in (a), predict the efficiency of new computer for data size 12(gigabytes). Does it possible to predict efficiency for data size of 30 gigabytes? Discuss.

Section B

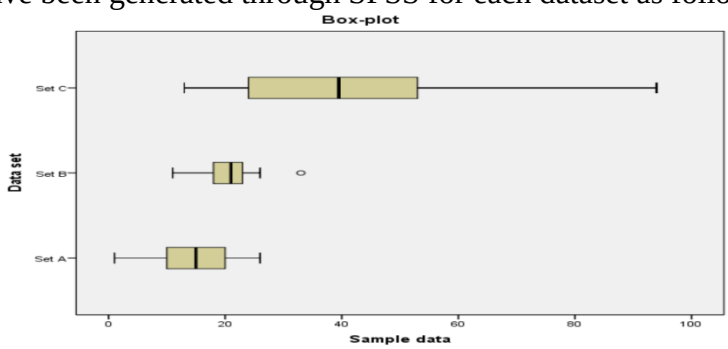
Attempt any eight questions. (8 × 5 = 40)

4. Explain the role of statistics in computer science and information technology.

5. Following table presents some descriptive statistics computed from three different independent sample dataset(X).

Data	Sample size (n)	$\sum_{i=1}^{30} X_i$	Minimum	Q1	Median	Q3	Maximum	$\sum_{i=1}^{30} (X_i - \bar{X})^2$
Data set A	30	439	1	10	15	20	26	1348.97
Data set B	30	625	11	18	21	23	33	540.17
Data set C	30	1239	13	24	39.5	53	94	12836.3

a) Compare sample mean and median, and explain about the shape of the data distribution for each dataset. Compare the variability of the three set of dataset.
Box-plots have been generated through SPSS for each dataset as follows.



b) Do these box-plots support your findings obtained in a) about the shape of the distribution? Explain.

6. A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%.

a) The inspector randomly picks 20 items from a shipment. What is the probability that there will be at least one defective item among these 20?
b) Suppose that the retailer receives 10 shipments in a month and the inspector randomly tests 20 devices per shipment. What is the probability that there will be exactly 3 shipments each containing at least one defective device among the 20 that are selected and tested from the shipment?

7. Messages arrive at an electronic message center at random times, with an average of 9 messages per hour.

a) What is the probability of receiving at least five messages during the next hour?
b) What is the probability of receiving exactly seven messages during the next hour?

8. The time, in minutes, it takes to reboot a certain system is a continuous variable with the density function:

$$f(x) = \begin{cases} C(10-x)^2, & 0 < x < 10 \\ 0, & \text{Otherwise} \end{cases}$$

Compute C, and then compute the probability that it takes between 1 and 2 minutes to reboot.

9. Following data represent the *preference of 10 students studying B.Sc .(CSIT) towards two brands of computers namely DELL and HP.*

Computer	**Student preference**									
DELL	5	2	9	8	1	10	3	4	6	7
HP	10	5	1	3	8	6	2	7	9	4

Apply appropriate statistical tool to measure whether the brand preference is correlated. Also interpret your result.

10. Define exponential distribution with parameter λ. The time required to reach to the printer after ordering in the computer follows exponential distribution at an average rate of 3 jobs per hour.

a) What is the expected time between jobs?
b) What is the probability that the next job is sent within 5 minutes?

11. The lifetime of a certain electronic component is a normal random variate with the expectation of 5000 hours and a standard deviation of 100 hours. Compute the probabilities under the following conditions

a) Lifetime of components is less than 5012 hours
b) Lifetime of components between 4000 to 6000 hours
c) Lifetime of components more than 7000 hours

12. Write short notes on the following.
    a) Sampling error and non-sampling error
    b) Conditional probability